# MATH 371 COMPUTING PROJECT 1

## SEPTEMBER 2, 2026

The object of this project is to apply Aitken's $\Delta^2$ method to accelerate the convergence of a sequence.

The sequence

$$\begin{cases} p_{n+1} &= \sqrt{3+p_n},\ n = 0, 1, 2, \ldots \\ p_0 &= 1, \end{cases}$$

can be shown to converge to $p = \dfrac{1+\sqrt{13}}{2}$.

**Part I: Theory/Analysis**

(i) Show that the sequence $\{p_n\}$ satisfies $1 \leq p_n \leq 3$

(ii) Show that $\{p_n\}$ is increasing. Conclude that it converges to a limit $p \leq 3$.

(iii) Show that the limit is indeed given by $p = \dfrac{1+\sqrt{13}}{2}$.

(iv) Compare $p_{n+1} - p$ to $p_n - p$ and evaluate the asymptotic constant $\lambda$

**Part II: Numerics**

(1) Generate the sequence $p_0, \ldots, p_{10}$

(2) Generate the sequence $\left| \dfrac{p_{n+1} - p}{p_n - p} \right|$, $n = 0, \ldots, 9$

(3) Generate the sequence $\hat{p}_n = p_n - \dfrac{(p_{n+1} - p_n)^2}{p_{n+2} - 2p_{n+1} + p_n}$, $n = 0, \ldots, 8$ using Aitken's formula.

(4) Generate the sequence $\left| \dfrac{\hat{p}_{n+1} - p}{\hat{p}_n - p} \right|$, $n = 0, \ldots, 7$

Use double precision in your calculations.
Place the computed numbers in the indicated columns as in the table.
Show 16 digits in columns 2 and 4 but 6 digits in columns 3, 5 and 6.
Leave the dashed cells empty.

| $n$ | $p_n$ | $\left\lvert\frac{p_{n+1}-p}{p_n-p}\right\rvert$ | $\hat{p}_n$ | $\left\lvert\frac{\hat{p}_{n+1}-p}{\hat{p}_n-p}\right\rvert$ | $\left\lvert\frac{\hat{p}_n-p}{p_n-p}\right\rvert$ |
|---|---|---|---|---|---|
| 0 | | | | | |
| 1 | | | | | |
| 2 | | | | | |
| 3 | | | | | |
| 4 | | | | | |
| 5 | | | | | |
| 6 | | | | | |
| 7 | | | | | |
| 8 | | | | —— | |
| 9 | | | —— | —— | —— |
| 10 | | —— | —— | —— | —— |

Answer the following questions

(i) What is the value of the asymptotic convergence rate $\lambda$ for each of the two sequences $p_n$ and $\hat{p}_n$?

(ii) Do the rates indicate that the sequence $\{\hat{p}_n\}$ converges faster than $p_n$?

(iii) Would you be correct to conclude that $\{\hat{p}_n\}$ converges superlinearly?